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Combined four-way valve, air conditioning system and vehicle

Abstract

Disclosed are a combined four-way valve, an air conditioning system, and a vehicle. The combined four-way valve includes a valve seat, a first control valve, a second control valve, a third control valve and a fourth control valve, where the valve seat is internally provided with an input port, an output port, a first opening, a second opening, and a plurality of valve cavities which are arranged at intervals and are arranged corresponding to the control valves on a one-to-one correspondence manner; the plurality of valve cavities include a first valve cavity, a second valve cavity, a third valve cavity and a fourth valve cavity; the valve cavities are in communication with one another by channels; the valve cavities are provided with inlets and outlets, and are in communication with flow channels in the valve seat.

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Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS

(1) The present disclosure is a national stage application of International Patent Application No. PCT/CN2022/116855, which is filed on Sep. 2, 2022. The International Patent Application claims the priority of Chinese Application No. 202111040533.9, filed in the Chinese Patent Office on Sep. 6, 2021, and entitled "Combined Four-way Valve, Air Conditioning System and Vehicle", claims the priority of Chinese Application No. 202122144603.7, filed in the Chinese Patent Office on Sep. 6, 2021, and entitled "Combined Four-way Valve, Air Conditioning System and Vehicle", and claims the priority of Chinese Application No. 202122143373.2, filed in the Chinese Patent Office on Sep. 6, 2021, and entitled "Integrated Control Multi-way Valve and Air Conditioning System", the entire contents of which are herein incorporated by reference.

TECHNICAL FIELD

(2) The present disclosure relates to a technical field of control valves, and more particularly to a combined four-way valve, an air conditioning system, and a vehicle.

BACKGROUND

(3) A household air conditioning system is usually provided with a four-way valve, and a flow direction of a refrigerant can be changed by actions of the four-way valve, thereby switching refrigeration and heating. The existing four-way valve achieves a flow path switching function by means of moving of one sliding block. However, the sliding block has a poor stability, and normal functions will be affected in case of moving of the sliding block in a use process. Therefore, the existing four-way valve is relatively applicable to a stable working environment, for example, in a fixed building. There is relatively great vibration in environments such as traveling vehicles, and thus the existing four-way valve cannot adapt to such environments. It is necessary to design a new four-way valve so as to adapt to the unstable working environments such as the vehicles.

SUMMARY

(4) The present disclosure provides a combined four-way valve, an air conditioning system, and a vehicle, so as to adapt to unstable working environments such as vehicles.

(5) In order to achieve the above purpose, according to one aspect of the present disclosure, some embodiments of the present disclosure provide the combined four-way valve, including: a valve seat, where the valve seat is internally provided with an input port, an output port, a first opening, a second opening, and a plurality of valve cavities which are arranged at intervals; each of the valve cavities is provided with an inlet and an outlet; the plurality of valve cavities include a first valve cavity, a second valve cavity, a third valve cavity and a fourth valve cavity; an inlet of the first valve cavity is in communication with the input port, an outlet of the first valve cavity is in communication with an outlet of the fourth valve cavity, the outlet of the fourth valve cavity is in communication with the second opening, an inlet of the second valve cavity is in communication with an inlet of the first valve cavity, an outlet of the second valve cavity is in communication with an outlet of the third valve cavity, the outlet of the third valve cavity is in communication with the first opening, an inlet of the third valve cavity is in communication with an inlet of the fourth valve cavity, and the inlet of the fourth valve cavity is in communication with the output port; a first control valve which is arranged in the first valve cavity in an openable and closable manner so as to control the inlet and the outlet of the first valve cavity to be in communication with or be disconnected from each other; a second control valve which is arranged in the second valve cavity

in an openable and closable manner so as to control the inlet and the outlet of the second valve cavity to be in communication with or be disconnected from each other; a third control valve which is arranged in the third valve cavity in an openable and closable manner so as to control the inlet and the outlet of the third valve cavity to be in communication with or be disconnected from each other; and a fourth control valve which is arranged in the fourth valve cavity in an openable and closable manner so as to control the inlet and the outlet of the fourth valve cavity to be in communication with or be disconnected from each other; wherein the combined four-way valve is capable of being switched to any one of a first operating state and a second operating state; under the circumstance that the combined four-way valve is in the first operating state, the second control valve and the fourth control valve are opened, and the first control valve and the third control valve are closed, such that the input port is in communication with the first opening, and the second opening is in communication with the output port; and under the circumstance that the combined four-way valve is in the second operating state, the second control valve and the fourth control valve are closed, and the first control valve and the third control valve are opened, such that the input port is in communication with the second opening, and the first opening is in communication with the output port.
(6) In some embodiments, the valve seat is further internally provided with a first channel, a second channel, a third channel and a fourth channel; the inlet of the first valve cavity is in communication with the inlet of the second valve cavity by the first channel; the inlet of the third valve cavity is in communication with the inlet of the fourth valve cavity by the second channel; the outlet of the first valve cavity is in communication with the outlet of the fourth valve cavity by the third channel; and the outlet of the second valve cavity is in communication with the outlet of the third valve cavity by the fourth channel.
(7) In some embodiments, the first channel and the input port are arranged coaxially, and the second channel and the output port are arranged coaxially; and the second valve cavity is located on a side of the first valve cavity facing away from the input port, and the third valve cavity is located on a side of the fourth valve cavity facing away from the output port.
(8) In some embodiments, the third channel and the second opening are arranged coaxially, and the fourth channel and the first opening are arranged coaxially; and the first valve cavity is located on a side of the fourth valve cavity facing away from the second opening, and the second valve cavity is located on a side of the third valve cavity facing away from the first opening.
(9) In some embodiments, the first channel, the second channel, the third channel and the fourth channel are blind holes.
(10) In some embodiments, depth directions of the plurality of valve cavities are the same direction, and in the depth directions of the plurality of valve cavities, the valve seat is provided with a first region and a second region which are spaced apart from each other; and the input port, the output port and inlets of the plurality of valve cavities are located in the first region, and the first opening, the second opening and outlets of the plurality of valve cavities are located in the second region.
(11) In some embodiments, an axis of the input port is parallel to an axis of the output port, an axis of the first opening is parallel to an axis of the second opening, the axis of the input port is perpendicular to the axis of the first opening, the axis of the input port is perpendicular to the depth directions of the plurality of valve cavities, and the axis of the first opening is perpendicular to the depth directions of the plurality of valve cavities.
(12) In some embodiments, the first control valve, the second control valve, the third control valve and the fourth control valve are electronic expansion valves; and the first control valve, the second control valve, the third control valve and the fourth control valve are detachably arranged on the valve seat.
(13) In some embodiments, the combined four-way valve further includes a control portion, where the control portion includes a protective box, and a circuit board and a plurality of coils which are

arranged in the protective box; the protective box is connected with the valve seat; the plurality of coils are electrically connected with the plurality of control valves in a one-to-one correspondence manner; and the plurality of coils are electrically connected to the circuit board.

(14) In some embodiments, the circuit board is provided with an integrated module and a conversion circuit module, the integrated module is connected with the conversion circuit module, and the plurality of coils are connected to the conversion circuit module.

(15) In some embodiments, the protective box includes a box body and a cover plate which are connected to each other, the box body is provided with an accommodating cavity, the circuit board is located in the accommodating cavity, and the plurality of coils are fixed in the accommodating cavity.

(16) In some embodiments, the circuit board is provided with a plurality of avoidance holes, and the plurality of coils penetrate into the plurality of avoidance holes in a one-to-one correspondence manner.

(17) In some embodiments, the combined four-way valve further includes a fixing member, where the protective box is connected with the valve seat by the fixing member.

(18) In some embodiments, the protective box is provided with a plurality of convex columns; the fixing member includes an annular member and a connecting member which are connected to each other; the annular member surrounds the plurality of control valves; the annular member is provided with a plurality of fixing holes; the plurality of convex columns penetrate into the plurality of fixing holes in a one-to-one correspondence manner; and the connecting member is connected with the valve seat by fasteners.

(19) According to another aspect of the present disclosure, the air conditioning system is provided. The air conditioning system includes a heat exchanger and the above combined four-way valve, where the heat exchanger is provided with a first refrigerant port and a second refrigerant port, the first refrigerant port is in communication with the first opening of the combined four-way valve, and the second refrigerant port is in communication with the second opening of the combined four-way valve.

(20) According to another aspect of the present disclosure, the vehicle is provided. The vehicle includes the above air conditioning system.

(21) By applying the technical solution of the present disclosure, the combined four-way valve is provided. The combined four-way valve includes the valve seat, the first control valve, the second control valve, the third control valve and the fourth control valve, where the valve seat is internally provided with the input port, the output port, the first opening, the second opening, and the plurality of valve cavities which are arranged at intervals; each of the valve cavities is provided with the inlet and the outlet; the plurality of valve cavities include the first valve cavity, the second valve cavity, the third valve cavity and the fourth valve cavity; the first control valve controls the inlet and the outlet of the first valve cavity to be in communication with or be disconnected from each other; the second control valve controls the inlet and the outlet of the second valve cavity to be in communication with or be disconnected from each other; the third control valve controls the inlet and the outlet of the third valve cavity to be in communication with or be disconnected from each other; the fourth control valve controls the inlet and the outlet of the fourth valve cavity to be in communication with or be disconnected from each other; where the combined four-way valve is capable of being switched to any one of the first operating state and the second operating state; under the circumstance that the combined four-way valve is in the first operating state, the second control valve and the fourth control valve are opened, and the first control valve and the third control valve are closed, such that the input port is in communication with the first opening, and the second opening is in communication with the output port; and under the circumstance that the combined four-way valve is in the second operating state, the second control valve and the fourth control valve are closed, and the first control valve and the third control valve are opened, such that the input port is in communication with the second opening, and the first opening is in

communication with the output port. In this solution, the plurality of control valves are used for respectively controlling the corresponding valve cavities to be opened and closed, and flow paths can be switched by means of opening and closing actions of the different control valves. This solution replaces the manner of using a sliding block in existing four-way valves, such that the problem of the sliding block being easy to move in a vibrating environment is avoided. In addition, the combined four-way valve in this solution switches the flow paths by means of cooperation of the plurality of control valves and can change a flow direction of a refrigerant, for example, the first operating state is applicable to a refrigeration condition, and the second operating state is applicable to a heating condition. The combined four-way valve provided in this solution has the high stability, and can adapt to unstable working environments such as vehicles, therefore ensuring reliable operation.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Drawings constituting a portion of the present disclosure are used for providing a further understanding to the present disclosure. Schematic embodiments of the present disclosure and descriptions thereof are intended to explain the present disclosure, and should not be construed to unduly limit the present disclosure. In the drawings:

(2) FIG. **1** shows a structural schematic diagram of a combined four-way valve provided in an embodiment of the present disclosure;

(3) FIG. **2** shows a top view of a valve seat in FIG. **1**;

(4) FIG. **3** shows a sectional view of the valve seat in FIG. **2** at position A-A;

(5) FIG. **4** shows a sectional view of the valve seat in FIG. **2** at position C-C;

(6) FIG. **5** shows a sectional view of the valve seat in FIG. **2** at position B-B;

(7) FIG. **6** shows a sectional view of the valve seat in FIG. **2** at position D-D;

(8) FIG. **7** shows a structural schematic diagram of a combined four-way valve provided in another embodiment of the present disclosure;

(9) FIG. **8** shows a schematic diagram of a control portion and a fixing member in FIG. **7**;

(10) FIG. **9** shows a schematic diagram of an air conditioning system provided in an embodiment of the present disclosure;

(11) FIG. **10** shows a schematic diagram of a combined four-way valve in the air conditioning system in FIG. **9** in a first state; and

(12) FIG. **11** shows a schematic diagram of the combined four-way valve in the air conditioning system in FIG. **9** in a second state.

(13) The above drawings include the following reference numerals: **10**. valve seat; **11**. input port; **12**. output port; **13**. first opening; **14**. second opening; **21**. first valve cavity; **22**. second valve cavity; **23**. third valve cavity; **24**. fourth valve cavity; **25**. first channel; **26**. second channel; **27**. third channel; **28**. fourth channel; **30**. first control valve; **40**. second control valve; **50**. third control valve; **60**. fourth control valve; **70**. heat exchanger; **80**. control portion; **81**. protective box; **811**. box body; **812**. cover plate; **813**. boss; **82**. circuit board; **821**. avoidance hole; **83**. coil; **90**. fixing member; **91**. annular member; and **92**. connecting member.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(14) The technical solutions in embodiments of the present disclosure will be described clearly and completely below in conjunction with the accompanying drawings in the embodiments of the present disclosure. Obviously, the described embodiments are only part of embodiments of the present disclosure, not all of them. Actually, the following description of at least one exemplary embodiment is only for the illustrative purpose, and shall not be construed as any limitation on the present disclosure or the disclosure and use thereof. On the basis of the embodiments of the present

disclosure, all other embodiments obtained by those of ordinary skill in the art without involving any inventive effort should fall within the scope of protection of the present disclosure.
(15) As shown in FIGS. **1-6**, one embodiment of the present disclosure provides a combined four-way valve, including: a valve seat **10**, where the valve seat **10** is internally provided with an input port **11**, an output port **12**, a first opening **13**, a second opening **14**, and a plurality of valve cavities which are arranged at intervals; each of the valve cavities is provided with an inlet and an outlet; the plurality of valve cavities include a first valve cavity **21**, a second valve cavity **22**, a third valve cavity **23** and a fourth valve cavity **24**; an inlet of the first valve cavity **21** is in communication with the input port **11**, an outlet of the first valve cavity **21** is in communication with an outlet of the fourth valve cavity **24**, the outlet of the fourth valve cavity **24** is in communication with the second opening **14**, an inlet of the second valve cavity **22** is in communication with the inlet of the first valve cavity **21**, an outlet of the second valve cavity **22** is in communication with an outlet of the third valve cavity **23**, the outlet of the third valve cavity **23** is in communication with the first opening **13**, an inlet of the third valve cavity **23** is in communication with an inlet of the fourth valve cavity **24**, and the inlet of the fourth valve cavity **24** is in communication with the output port **12**; a first control valve **30** which is arranged in the first valve cavity **21** in an openable and closable manner so as to control the inlet and the outlet of the first valve cavity **21** to be in communication with or be disconnected from each other; a second control valve **40** which is arranged in the second valve cavity **22** in an openable and closable manner, so as to control the inlet and the outlet of the second valve cavity **22** to be in communication with or be disconnected from each other; a third control valve **50** which is arranged in the third valve cavity **23** in an openable and closable manner so as to control the inlet and the outlet of the third valve cavity **23** to be in communication with or be disconnected from each other; and a fourth control valve **60** which is arranged in the fourth valve cavity **24** in an openable and closable manner so as to control the inlet and the outlet of the fourth valve cavity **24** to be in communication with or be disconnected from each other; wherein the combined four-way valve is capable of being switched to any one of a first operating state and a second operating state; under the circumstance that the combined four-way valve is in the first operating state, the second control valve **40** and the fourth control valve **60** are opened, and the first control valve **30** and the third control valve **50** are closed, such that the input port **11** is in communication with the first opening **13**, and the second opening **14** is in communication with the output port **12**; and under the circumstance that the combined four-way valve is in the second operating state, the second control valve **40** and the fourth control valve **60** are closed, and the first control valve **30** and the third control valve **50** are opened, such that the input port **11** is in communication with the second opening **14**, and the first opening **13** is in communication with the output port **12**.
(16) In this solution, the plurality of control valves are used for respectively controlling corresponding valve cavities to be opened and closed, and flow paths can be switched by opening and closing actions of the different control valves. This solution replaces the manner of using a sliding block in existing four-way valves, such that a problem of the sliding block being prone to move in a vibrating environment is avoided. In addition, the combined four-way valve in this solution switches the flow paths by means of cooperation of the plurality of control valves and can change a flow direction of a refrigerant, for example, the first operating state is applicable to a refrigeration condition, and the second operating state is applicable to a heating condition. The combined four-way valve provided in this solution has a high stability, and is able to adapt to unstable working environments such as vehicles, therefore ensuring reliable operation.
(17) In this solution, the valve seat **10** is internally provided with the plurality of valve cavities at intervals, each of the valve cavities is internally provided with the inlet and the outlet, the input port **11**, the output port **12**, the first opening **13** and the second opening **14** in the valve seat **10** are in communication with the inlets or the outlets of the first valve cavity **21**, the second valve cavity **22**, the third valve cavity **23** and the fourth valve cavity **24** to form flow channels in the combined

four-way valve, the flow channels are simpler than those of existing valve seats **10**, and the communication or disconnection of the inlets and the outlets in the valve cavities is controlled by the first control valve **30**, the second control valve **40**, the third control valve **50** and the fourth control valve **60** which are correspondingly arranged in the valve cavities in the openable and closable manner, thereby achieving the different operating states.
(18) In some embodiments, the valve seat **10** is further internally provided with a first channel **25**, a second channel **26**, a third channel **27** and a fourth channel **28**; the inlet of the first valve cavity **21** is in communication with the inlet of the second valve cavity **22** by the first channel **25**; the inlet of the third valve cavity **23** is in communication with the inlet of the fourth valve cavity **24** by the second channel **26**; the outlet of the first valve cavity **21** is in communication with the outlet of the fourth valve cavity **24** by the third channel **27**; and the outlet of the second valve cavity **22** is in communication with the outlet of the third valve cavity **23** by the fourth channel **28**. The valve cavities are connected by the channels, which greatly simplifies the flow channels, and facilitates machining of the flow channels.
(19) In some embodiments, the first channel **25** and the input port **11** are arranged coaxially, and the second channel **26** and the output port **12** are arranged coaxially; and the second valve cavity **22** is located on a side of the first valve cavity **21** facing away from the input port **11**, and the third valve cavity **23** is located on a side of the fourth valve cavity **24** facing away from the output port **12**. In this way, it is ensured that portions to be machined are located on the same path, such that the channels can be machined and formed with the machining of the input port **11**, the output port **12**, the first opening **13** and the second opening **14**, and machining steps for the valve seat **10** are simplified.
(20) In this solution, the flow channels in the valve seat **10** are straight flow channels, there is no flow channel needing complex machining, and thus the flow channels in the valve seat **10** include: a flow channel where the input port **11** is located, a flow channel where the output port **12** is located, a flow channel where the first opening **13** is located, and a flow channel where the second opening **14** is located; in addition, the first valve cavity **21**, the second valve cavity **22**, the third valve cavity **23** and the fourth valve cavity **24** are relatively easy to machine and are not required to be machined into through holes during machining, thereby avoiding the situation that plugs are needed to block redundant openings.
(21) In some embodiments, axes of an inlet and an outlet of each control valve are perpendicular to an axis of the control valve, the axis of the inlet of the control valve is perpendicular to the axis of the outlet of the control valve, the each control valve is arranged in a corresponding valve cavity in the openable and closable manner, the inlet of the control valve and the flow channel where the input port **11** or the output port **12** is located are arranged coaxially, and the outlet of the control valve and the flow channel where the first opening **13** or the second opening **14** is located are arranged coaxially. In this solution, each control valve controls the communication or disconnection of the internal flow channels by controlling the opening and closing of a region between the inlet and the outlet.
(22) In some embodiments, the first channel **25**, the second channel **26**, the third channel **27** and the fourth channel **28** are blind holes. Different from the configuration manner of blocking one end of each through hole in a related art, this solution avoids fluid leakage caused by using blocking plugs; in addition, during machining of the blind holes in this solution, the situation that machining burrs affect mounting of the first control valve **30**, the second control valve **40**, the third control valve **50** and the fourth control valve **60** can be avoided.
(23) In this solution, depth directions of the plurality of valve cavities are the same direction, and in the depth directions of the plurality of valve cavities, the valve seat **10** is provided with a first region and a second region which are spaced apart from each other; and the input port **11**, the output port **12** and the inlets of the plurality of valve cavities are located in the first region, and the first opening **13**, the second opening **14** and the outlets of the plurality of valve cavities are located

in the second region. Thus, the arrangement of the openings is facilitated, the lengths of the flow channels are shortened, the compactness of the valve seat **10** is improved, and the size of the valve seat **10** is reduced.
(24) In some embodiments, an axis of the input port **11** is parallel to an axis of the output port **12**, an axis of the first opening **13** is parallel to an axis of the second opening **14**, the axis of the input port **11** is perpendicular to the axis of the first opening **13**, the axis of the input port **11** is perpendicular to the depth directions of the plurality of valve cavities, and the axis of the first opening **13** is perpendicular to the depth directions of the plurality of valve cavities.
(25) In this way, the design of the flow channels in the valve seat **10** is simplified, design and fitting can be performed according to relative positions of the input port **11**, the output port **12**, the first opening **13**, the second opening **14** and all the valve cavities, and positions of the flow channels are determined in space, thereby further simplifying the flow channels, facilitating machining, and reducing the production cost.
(26) In some embodiments, the first control valve **30**, the second control valve **40**, the third control valve **50** and the fourth control valve **60** are electronic expansion valves; and the first control valve **30**, the second control valve **40**, the third control valve **50** and the fourth control valve **60** are detachably arranged on the valve seat. With the adoption of the electronic expansion valves, a speed of the combined four-way valve for a thermal reaction is increased, the evaporation speed is stabler, meanwhile, the adverse effect caused by the excessively high temperature is also prevented, and the reliability is improved. In addition, the electronic expansion valves not only have the function of opening and closing valve ports, but also have the function of adjusting the flow, and thus the combined four-way valve also has the effect of adjusting the flow, which enriches the functions of the combined four-way valve.
(27) In this embodiment, the plurality of control valves, including the first control valve **30**, the second control valve **40**, the third control valve **50** and the fourth control valve **60**, of the combined four-way valve are respectively provided with coils and are separately controlled.
(28) As shown in FIGS. **7-8**, another embodiment of the present disclosure provides a combined four-way valve, and different from the above embodiment, the combined four-way valve further includes a control portion **80**, where the control portion **80** includes a protective box **81**, a circuit board **82** and a plurality of coils **83**, the circuit board **82** and the plurality of coils **83** are arranged in the protective box **81**; the protective box **81** is connected with the valve seat **10**; the plurality of coils **83** are electrically connected to the plurality of control valves in a one-to-one correspondence manner; and the plurality of coils **83** are electrically connected with the circuit board **82**.
(29) With the adoption of this solution, the plurality of coils for driving the plurality of control valves to act are arranged in a concentrated manner in the protective box **81** and are connected with the circuit board **82**, thus the integrated control portion **80** is formed, the wiring number is reduced, all that is needed is to connect the control portion **80** and the valve seat **10** to the corresponding control valves during assembly, and the assembly efficiency is improved; in addition, cooperative control can be performed on the plurality of control valves by the integrated control portion **80** and replaces the respective and separate control manner, such that the convenience in control is improved.
(30) At least part of the structure of each control valve is mounted in the corresponding valve cavity, and part of the structure of the control valve can protrude out of the valve cavity, such that fitting of the coils **83** in the protective box **81** is more convenient, an occupied space of the coils **83** in the entire protective box **81** is reduced, and the structure is more compact.
(31) In some embodiments, the circuit board **82** is provided with an integrated module and a conversion circuit module, the integrated module is connected with the conversion circuit module, and the plurality of coils **83** are connected with the conversion circuit module. The integrated module is used for centralized control, and the conversion circuit module is used for converting signals, thereby achieving a purpose of cooperative control.

(32) In this solution, the integrated module is used, which greatly simplifies the design and mounting of an entire circuit, and makes the circuit looks simpler; in addition, the integrated module has a certain advantage in the aspect of energy consumption, is low in power consumption and small in size, has certain economic applicability, and has the advantage of high reliability, thereby improving a working reliability of the entire circuit.
(33) In this solution, the conversion circuit module is used, and the conversion circuit module also has the advantages of simple design, high reliability and flexible design; in addition, the conversion circuit module also has the advantages of high power, high efficiency and easiness in maintenance, has multiple input and output choices, and is easier and safer to debug.
(34) In this embodiment, the protective box **81** includes two portions, namely, a box body **811** and a cover plate **812**, structures such as the circuit board **82** and the like are arranged in an accommodating cavity of the box body **811**, and thus a certain protection effect is achieved on the entire structure; and the plurality of coils **83** are fixed in the accommodating cavity by a pouring technology or an injection molding technology, which limits the positions of the coils **83** in the accommodating cavity, and can also prevent, to a certain extent, the coils **83** from moving. The box body **811** can be connected to the cover plate **812** by fusion, riveting, snap-fitting or welding, which is convenient, safe and easy to operate.
(35) In some embodiments, the circuit board **82** is provided with four avoidance holes **821**, and the plurality of coils **83** penetrate into the plurality of avoidance holes **821** in a one-to-one correspondence manner. In this embodiment, the plurality of coils **83** are inserted, in the one-to-one correspondence manner, into the plurality of avoidance holes **821** formed in the circuit board **82**, such that the coils are fitted with the circuit board **82**, a height of the accommodating cavity in the protective box **81** and a connection height between the control portion **80** and the valve seat are shortened, and a size of the entire structure is reduced.
(36) Specifically, the combined four-way valve further includes a fixing member **90**, where the protective box **81** is connected with the valve seat **10** by the fixing member **90**.
(37) In this embodiment, the combined four-way valve is designed in such a structure that the protective box **81** is connected with the valve seat **10** by the fixing member **90**, the reliable connection between the protective box and the valve seat is facilitated, and the structure stability is ensured.
(38) In some embodiments, the protective box **81** is provided with a plurality of convex columns **813**; the fixing member **90** includes an annular member **91** and a connecting member **92** which are connected to each other; the annular member **91** surrounds the plurality of control valves; the annular member **91** is provided with a plurality of fixing holes; the plurality of convex columns **813** penetrate into the plurality of fixing holes in a one-to-one correspondence manner; and the connecting member **92** is connected with the valve seat **10** by fasteners.
(39) In this embodiment, a bottom end of the protective box **81** is provided with the plurality of convex columns **813**, which are fitted with the fixing holes in the annular member **91** included by the fixing member **90**, thereby achieving fixed connection. For example, the protective box **81** and the convex columns **813** are made of plastic, the fixing member **90** is made of metal, and after the convex columns **813** penetrate through the annular member **91**, the portions of the convex columns **813** penetrating through the fixing holes of the annular member **91** are melted and attached, by means of welding, to surfaces of the fixing holes, such that protruding portions of the convex columns **813** become new convex columns that can cover the fixing holes; and after the new convex columns are cooled, the diameters of the protruding portions are greater than those of the fixing holes, snap-fitting of the protective box **81** and the fixing member **90** is achieved. A side face of the fixing member **90** is provided with the connecting member **92**, which is connected to the annular member **91** and is arranged perpendicular to the annular member **91**, the connecting member is provided with two holes that correspond to holes in the valve seat **10**, and the connecting member is fixed to the valve seat by the fasteners such as bolts.

(40) Specifically, in this embodiment, the four control valves are controlled as a whole by an LIN bus, where the second control valve **40** and the fourth control valve **60** are opened and closed synchronously, and the first control valve **30** and the third control valve **50** are opened and closed synchronously.
(41) As shown in FIGS. **9-11**, another embodiment of the present disclosure provides an air conditioning system. The air conditioning system includes a heat exchanger **70** and the above combined four-way valve, where the heat exchanger **70** is provided with a first refrigerant port and a second refrigerant port, the first refrigerant port is in communication with the first opening **13** of the combined four-way valve, and the second refrigerant port is in communication with the second opening **14** of the combined four-way valve. In the air conditioning system, the plurality of control valves are used for respectively controlling the corresponding valve cavities to be opened and closed, and flow paths can be switched by means of opening and closing actions of the different control valves. This solution replaces the manner of using a sliding block in existing four-way valves, such that the problem of the sliding block being prone to moving in a vibrating environment is avoided, the reliability of the air conditioning system is improved, and the air conditioning system can adapt to unstable working environments such as vehicles, therefore ensuring reliable operation.
(42) Some embodiments of the present disclosure further provide the vehicle. The vehicle includes the above air conditioning system.
(43) The technical solutions in the present disclosure can bring the following beneficial effects: 1. the plurality of control valves are used for respectively controlling the corresponding valve cavities to be opened and closed, and the flow paths can be switched by means of the different opening and closing actions; 2. the manner of using the sliding block in the existing four-way valves is replaced, such that the problem that the sliding block is easy to move in the vibrating environment is avoided; and 3. the combined four-way valve has high stability, and can adapt to the unstable working environments such as the vehicles, therefore ensuring reliable operation.
(44) The above are some embodiments of the present disclosure and are not intended to limit the present disclosure, and various changes and modifications may be made on the present disclosure by those skilled in the art. Any modification, equivalent substitution, improvement, etc. made within the spirit and principles of the present disclosure should fall within the scope of protection of the present disclosure.

Claims

1. A combined four-way valve, comprising: a valve seat, wherein the valve seat is internally provided with an input port, an output port, a first opening, a second opening, and a plurality of valve cavities which are arranged at intervals; each of the valve cavities is provided with an inlet and an outlet; the plurality of valve cavities comprise a first valve cavity, a second valve cavity, a third valve cavity and a fourth valve cavity; an inlet of the first valve cavity is in communication with the input port, an outlet of the first valve cavity is in communication with an outlet of the fourth valve cavity, the outlet of the fourth valve cavity is in communication with the second opening, an inlet of the second valve cavity is in communication with the inlet of the first valve cavity, an outlet of the second valve cavity is in communication with an outlet of the third valve cavity, the outlet of the third valve cavity is in communication with the first opening, an inlet of the third valve cavity is in communication with an inlet of the fourth valve cavity, and the inlet of the fourth valve cavity is in communication with the output port; a first control valve arranged in the first valve cavity in an openable and closable manner, so as to control the inlet and the outlet of the first valve cavity to be in communication with or be disconnected from each other; a second control valve arranged in the second valve cavity in an openable and closable manner, so as to control the inlet and the outlet of the second valve cavity to be in communication with or be disconnected from

each other; a third control valve arranged in the third valve cavity in an openable and closable manner, so as to control the inlet and the outlet of the third valve cavity to be in communication with or be disconnected from each other; and a fourth control valve arranged in the fourth valve cavity in an openable and closable manner, so as to control the inlet and the outlet of the fourth valve cavity to be in communication with or be disconnected from each other; wherein the combined four-way valve is capable of being switched to any one of a first operating state and a second operating state; in response to that the combined four-way valve is in the first operating state, the second control valve and the fourth control valve are opened, and the first control valve and the third control valve are closed, such that the input port is in communication with the first opening, and the second opening is in communication with the output port; and in response to that the combined four-way valve is in the second operating state, the second control valve and the fourth control valve are closed, and the first control valve and the third control valve are opened, such that the input port is in communication with the second opening, and the first opening is in communication with the output port.

2. The combined four-way valve according to claim 1, wherein the valve seat is further internally provided with a first channel, a second channel, a third channel and a fourth channel; the inlet of the first valve cavity is in communication with the inlet of the second valve cavity by the first channel; the inlet of the third valve cavity is in communication with the inlet of the fourth valve cavity by the second channel; the outlet of the first valve cavity is in communication with the outlet of the fourth valve cavity by the third channel; and the outlet of the second valve cavity is in communication with the outlet of the third valve cavity by the fourth channel.

3. The combined four-way valve according to claim 2, wherein the first channel and the input port are arranged coaxially, and the second channel and the output port are arranged coaxially; and the second valve cavity is located on a side of the first valve cavity facing away from the input port, and the third valve cavity is located on a side of the fourth valve cavity facing away from the output port.

4. The combined four-way valve according to claim 2, wherein the third channel and the second opening are arranged coaxially, and the fourth channel and the first opening are arranged coaxially; and the first valve cavity is located on a side of the fourth valve cavity facing away from the second opening, and the second valve cavity is located on a side of the third valve cavity facing away from the first opening.

5. The combined four-way valve according to claim 2, wherein the first channel, the second channel, the third channel and the fourth channel are blind holes.

6. The combined four-way valve according to claim 1, wherein depth directions of the plurality of valve cavities are a same direction, and in the depth directions of the plurality of valve cavities, the valve seat is provided with a first region and a second region which are spaced apart from each other; and the input port, the output port and inlets of the plurality of valve cavities are located in the first region, and the first opening, the second opening and outlets of the plurality of valve cavities are located in the second region.

7. The combined four-way valve according to claim 6, wherein an axis of the input port is parallel to an axis of the output port, an axis of the first opening is parallel to an axis of the second opening, the axis of the input port is perpendicular to the axis of the first opening, the axis of the input port is perpendicular to the depth directions of the plurality of valve cavities, and the axis of the first opening is perpendicular to the depth directions of the plurality of valve cavities.

8. The combined four-way valve according to claim 1, wherein the first control valve, the second control valve, the third control valve and the fourth control valve are electronic expansion valves; and the first control valve, the second control valve, the third control valve and the fourth control valve are detachably arranged on the valve seat.

9. The combined four-way valve according to claim 1, further comprising a control portion, wherein the control portion comprises a protective box, and a circuit board and a plurality of coils

which are arranged in the protective box; the protective box is connected with the valve seat; the plurality of coils are electrically connected to a plurality of control valves in a one-to-one correspondence manner; and the plurality of coils are electrically connected to the circuit board.

10. The combined four-way valve according to claim 9, wherein the circuit board is provided with an integrated module and a conversion circuit module, the integrated module is connected with the conversion circuit module, and the plurality of coils are connected to the conversion circuit module.

11. The combined four-way valve according to claim 9, wherein the protective box comprises a box body and a cover plate which are connected to each other, the box body is provided with an accommodating cavity, the circuit board is located in the accommodating cavity, and the plurality of coils are fixed in the accommodating cavity.

12. The combined four-way valve according to claim 9, wherein the circuit board is provided with a plurality of avoidance holes, and the plurality of coils penetrate into the plurality of avoidance holes in a one-to-one correspondence manner.

13. The combined four-way valve according to claim 9, further comprising a fixing member, wherein the protective box is connected with the valve seat by the fixing member.

14. The combined four-way valve according to claim 13, wherein the protective box is provided with a plurality of convex columns; the fixing member comprises an annular member and a connecting member which are connected to each other; the annular member surrounds the plurality of control valves; the annular member is provided with a plurality of fixing holes; the plurality of convex columns penetrate into the plurality of fixing holes in a one-to-one correspondence manner; and the connecting member is connected with the valve seat by fasteners.

15. An air conditioning system, comprising a heat exchanger and the combined four-way valve according to claim 1, wherein the heat exchanger is provided with a first refrigerant port and a second refrigerant port, the first refrigerant port is in communication with the first opening of the combined four-way valve, and the second refrigerant port is in communication with the second opening of the combined four-way valve.

16. The air conditioning system according to claim 15, wherein the valve seat is further internally provided with a first channel, a second channel, a third channel and a fourth channel; the inlet of the first valve cavity is in communication with the inlet of the second valve cavity by the first channel; the inlet of the third valve cavity is in communication with the inlet of the fourth valve cavity by the second channel; the outlet of the first valve cavity is in communication with the outlet of the fourth valve cavity by the third channel; and the outlet of the second valve cavity is in communication with the outlet of the third valve cavity by the fourth channel.

17. The air conditioning system according to claim 16, wherein the first channel and the input port are arranged coaxially, and the second channel and the output port are arranged coaxially; and the second valve cavity is located on a side of the first valve cavity facing away from the input port, and the third valve cavity is located on a side of the fourth valve cavity facing away from the output port.

18. The air conditioning system according to claim 16, wherein the third channel and the second opening are arranged coaxially, and the fourth channel and the first opening are arranged coaxially; and the first valve cavity is located on a side of the fourth valve cavity facing away from the second opening, and the second valve cavity is located on a side of the third valve cavity facing away from the first opening.

19. The air conditioning system according to claim 16, wherein the first channel, the second channel, the third channel and the fourth channel are blind holes.

20. A vehicle, comprising the air conditioning system according to claim 15.